

Federico Melo Barrero

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EXPERIENCE

Canals AI

Dec 2024 – Present · 1 yr 10 mos

Coral Gables, FL, USA · Remote · Full-time

Software Engineer II · Parsing

Jul 2025 – Present · 1 yr 3 mos

- Owned parsing services for the live voice and inquiries agent systems, as part of a pipeline that processed 20000+ orders daily across 115+ wholesale distributors and 9000+ sales reps
- Built the parsing and review layers of a real-time voice service (WebSocket orchestration, speech-to-text, speaker diarization, concurrency control). Extracted ~21 products per call from live phone calls with 75%+ retained through rep review
- Integrated tools (web search, spec sheet retrieval, ERP lookup, order tracking) into an inquiries AI agent that addressed customer questions on orders, materials, and logistics. Achieved 97% of answers citing a source and 80% rated positive by reps
- Contributed to Canals' shared agent framework (Mastra), adding extended thinking, reasoning effort, image/PDF input, temperature, and tool concurrency. Migrated four Parsing agents off hand-rolled, unbounded loops onto bounded runs with step and token ceilings
- Co-built an autonomous agent (Claude Agent SDK) that polled tickets, debugged parsing failures, and shipped prompt-level fixes for review, re-parsing each order to verify its own fix
- One of 4 engineers on the 10-person Parsing team (~60 engineers total) holding merge approval rights on the main branch.

Software Engineer I · Parsing

Dec 2024 – Jul 2025 · 7 mos

- Maintained a parsing pipeline supporting 19 mimetypes with intricate formats (e.g., handwritten, struck-through text, blurry and fragmented photos, cross-references to other files) in a pipeline serving ~60 wholesale distributors processing 4,000+ orders daily
- Abstracted OCR pipeline from single provider to support multiple providers, decoupling the implementation from Textract and enabling format-specific OCR selection, with a cache layer serving ~60% of requests in 0.07s versus 3.6–18.7s uncached
- Led large-scale prompt refactors improving external QA pass rate from 85% to 89%. Involved: model selection, prompt engineering, cache handling, fine-tuning, and regression testing
- Built visual highlighting systems using DOM traversal, coordinate mapping, and string comparison to map parsed products to source documents. Reduced weekly highlighting errors by 2/3
- Joined as the company's 14th engineer.

Universidad de los Andes

Aug 2025 – Present · 1 yr 2 mos

Adjunct Professor · Algorithm Design

Fall 2026 · 2 mos

- Principal instructor for upper-division algorithm design course, delivering lectures and designing exams
- Topics: divide and conquer, dynamic programming, graph algorithms, greedy methods, NP-completeness, approximation and randomized algorithms

Adjunct Professor · Introduction to Programming

Fall 2025, Spring 2026 · 9 mos

- Principal instructor for introductory undergraduate course in procedural programming using Python
- Topics: programming fundamentals and data-processing libraries Pandas and Matplotlib

Universidad de los Andes

Bogotá, Colombia · On-site · Part-time

Full-Stack Developer · Vice Dean's Office of Student Affairs

Jan 2024 – Dec 2024 · 11 mos

- Designed and built full-stack student data analytics system serving 15,000+ students and 600+ faculty, replacing an external platform and saving ~\$100K USD annually
- Built parallelized data processing pipeline (Python, Polars) with complex filtering and aggregations, exposed through a unified REST API (FastAPI)
- Developed dynamic graphing, advanced filtering, report generation, and role-based dashboards in a React/TypeScript web app
- Built early-warning system for at-risk students (academic performance, dropout indicators), alerting counselors on 1,000+ critical cases
- Architected secure cross-VM communication using air-gapping and network isolation techniques to ensure compliance with student data protection requirements

- Caseware International Inc.**
Software Developer I · Notebook Ninjas, Data Analytics

Toronto, ON, Canada · Remote · Internship
Jan 2024 – Jul 2024 · 6 mos

 - Awarded June Team Award for impactful contributions as part of Data Analytics team
 - Developed Jupyter notebooks for financial and auditing data analysis
 - Optimized microservices (Java, TS) performance using AWS S3 and DynamoDB
 - Authored technical reports influencing team architecture decisions and best practices
- Universidad de los Andes**
Backend Developer · Engineering Faculty Research Center (CIFI)

Bogotá, Colombia · Hybrid · Part-time
Aug 2023 – Dec 2023 · 4 mos

 - Designed and implemented backend architecture for the Professors' Portfolio system (Java, Spring Boot), a web application for managing faculty academic and administrative information.
 - Built persistence, logic, and control layers with CRUD operations, custom logic, and RESTful APIs
 - Devised efficient data migration methods for unstructured faculty data
 - Collaborated on system architecture documentation and project planning roadmaps

EDUCATION

- Universidad de los Andes**
B.Sc. Systems and Computing Engineering, Summa Cum Laude

Bogotá, Colombia
2020 – 2024

 - Cumulative GPA: 4.92 / 5.00 — Highest GPA in major in 40 years; highest in Engineering in 16 years

AWARDS & HONORS

- Dejar Huella Award**
Uniandes internal impact award, for a peer-support platform serving 56,000+ students

Universidad de los Andes, 2026
- Award for Outstanding Saber Pro and TyT Scores 2024**
Colombia's mandatory national exit exam for graduating university students

Colombian Ministry of National Education, 2025
- Summa Cum Laude**
Highest undergraduate honor; top 1% cumulative GPA among graduates of the faculty

Universidad de los Andes, 2025
- Athletic Career Distinction**
Awarded to one graduating student university-wide for leadership and excellence in varsity athletics

Universidad de los Andes, 2025
- Ramón de Zubiría Award (2 times)**
Awarded annually to the top cumulative-GPA student in the major

Universidad de los Andes, 2023, 2024
- Semester Excellence Award (3 times)**
Awarded each semester to the top-GPA student in the major

Universidad de los Andes, Spring 2021, Fall 2021, Fall 2023

TECHNICAL EXPERIENCE

- AI Engineering & LLM Orchestration:** Anthropic / OpenAI / Gemini / Groq APIs, Claude Agent SDK, Mastra, MCP
- Backend & Systems Architecture:** Python, Java, TypeScript, FastAPI, Spring Boot, WebSocket Orchestration, Concurrency Control
- Data Engineering & Infrastructure:** Polars, PostgreSQL, Redis, AWS (RDS, ECS, S3, DynamoDB), Docker, GitHub CI/CD
- Product & Interface:** React, Next.js, Svelte, Plotly, ShadCN / MUI